

1. Scene & viewpoint filtering



Filter HM3D rooms
with smart-home devices



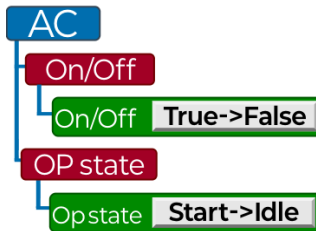
Select camera
position & angle



2. Target device & instruction generation



Target device & goal selection



Instruction generation

[Prompt]

You are a homeowner giving a natural voice
command to your smart home system... (etc.)



GPT-5 mini

[Query]

That humming is loud.
I've got a call. Silence it.

3. Speech & avatar synthesis

Reference voice



+

Query

That humming is loud.
I've got a call. Silence it.

=

Text to speech



Reference avatar



+

Pose skeleton



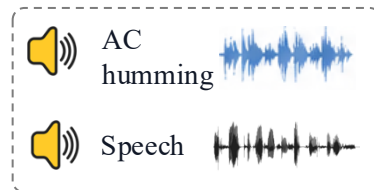
=

Avatar render

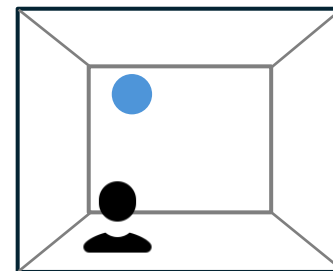


4. Audio-visual scene composition

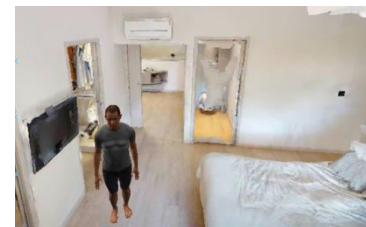
Object Sounds



Spatialize & mix



Final output



Visual Image + FOA audio